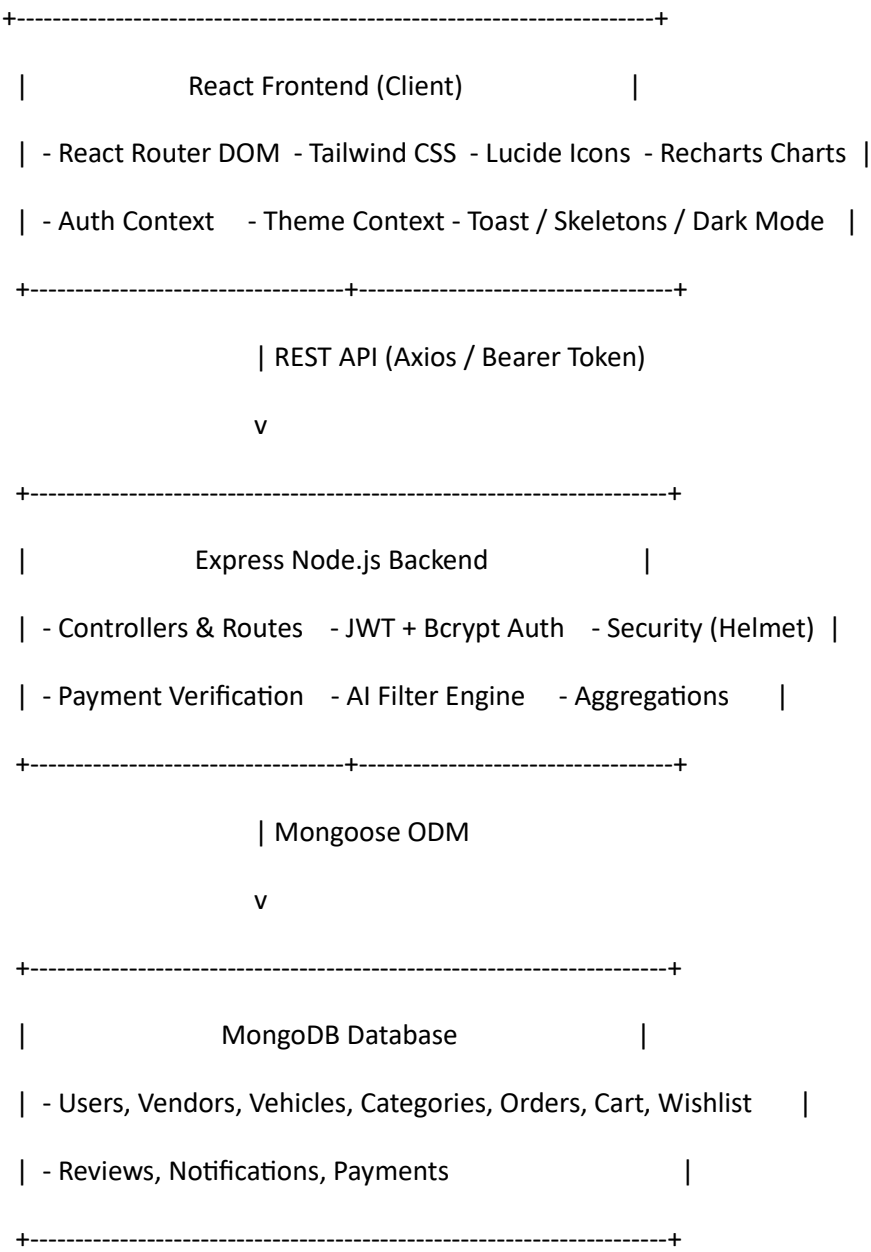


Implementation Plan - Production-Quality Electric Vehicle Marketplace (MERN)

Build a complete, production-ready, full-stack Electric Vehicle Marketplace using MongoDB, Express.js, React, Node.js, Tailwind CSS, Stripe, and AI recommendations.

Technical Architecture & Design Principles



User Review Required

IMPORTANT

- **Architecture:** Full MERN Stack with Express backend listening on port 5000 (or PORT env) and React Vite frontend on port 5173.
 - **Stripe & AI:** Works seamlessly out-of-the-box with mock/fallback mode when API keys (STRIPE_SECRET_KEY, AI_API_KEY) are not provided, while supporting real API integration when keys are present in .env.
 - **Roles & Permissions:** Strict backend RBAC enforcing CUSTOMER, VENDOR, and ADMIN access. Admin creation is secured via seed script or explicit secret key.
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Proposed Implementation Phases

Phase 1: Workspace Setup & Core Infrastructure

- Initialize backend/ (package.json, Express, CORS, Helmet, dotenv, Mongoose, JWT, bcryptjs, Stripe).
- Initialize frontend/ (Vite + React, Tailwind CSS, Lucide React, Recharts, Axios, React Router).
- Setup environment config files (.env, .env.example).

Phase 2: MongoDB Schemas & Data Models (backend/src/models/)

- User.js: name, email, password (hashed), phone, role (CUSTOMER, VENDOR, ADMIN), avatar, address, status.
- Vendor.js: userRef, businessName, description, logo, contactPhone, contactEmail, isVerified, rating, totalSales.
- Vehicle.js: title, brand, model, year, price, description, condition (NEW, USED), mileage, rangeKm, batteryCapacityKwh, chargingTimeHours, topSpeedKmh, seatingCapacity, bodyType, location, features (array), images (array), vendor (ref), status (DRAFT, PENDING_APPROVAL, APPROVED, REJECTED, SOLD), rejectionReason, ratingsAverage, ratingsQuantity.
- Category.js: name, slug, description, icon.
- Cart.js: user (ref), items (vehicle ref, priceAtAddition, quantity).
- Wishlist.js: user (ref), vehicles (array of Vehicle refs).
- Order.js: orderNumber, customer (ref), items (array of items with vehicle, price, vendor), shippingAddress, paymentInfo (id, status, method), pricing (subtotal, tax, deliveryFee, totalPrice), orderStatus (PENDING, CONFIRMED, PROCESSING, SHIPPED, OUT_FOR_DELIVERY, DELIVERED, CANCELLED), statusHistory.
- Review.js: vehicle (ref), user (ref), order (ref), rating (1-5), comment, createdAt.

- Notification.js: user (ref), title, message, type, read, link.
- Payment.js: order (ref), user (ref), stripePaymentIntentId, amount, currency, status.

Phase 3: Database Seeding & Mock Data (backend/src/seeder/)

- Seed database with realistic EVs: Tesla Model 3, Tesla Model Y, Tata Nexon EV, Tata Curvv EV, Mahindra XUV400, Hyundai Ioniq 5, MG ZS EV, Kia EV6, BMW iX1, Mercedes EQB.
- Seed customers, verified vendors, admin account, categories, and sample reviews.

Phase 4: Backend API Controllers, Services & Routes (backend/src/)

- Auth: /api/v1/auth/register, /login, /me, /profile
- Vehicles: /api/v1/vehicles (Search, Filter, Sort, Pagination, Natural Language Filter, Details)
- Cart & Wishlist: /api/v1/cart, /api/v1/wishlist
- Checkout & Payments: /api/v1/payments/create-intent, /verify-payment, Stripe Webhooks
- Orders: /api/v1/orders, /my-orders, /:id/track, /status-update
- Vendor: /api/v1/vendor/dashboard, /my-vehicles, /orders, /sales
- Admin: /api/v1/admin/dashboard, /users, /vendors, /vehicles/approvals, /analytics
- Calculators & AI: /api/v1/calculators, /api/v1/recommendations
- Reviews & Notifications: /api/v1/reviews, /api/v1/notifications

Phase 5: Frontend Design System & Theme Engine (frontend/src/)

- Dark & Light mode toggle with persistent state in localStorage & html class dark.
- Toast notification context & standard reusable UI components (Button, Card, Badge, Modal, Input, Select, Skeleton, EmptyState, ErrorState).

Phase 6: Core Customer Application Pages

- **Homepage:** Hero, Featured EVs (live DB data), Category Grid, EV Benefits, Interactive Calculators teaser, AI recommendation widget.
- **Explore Marketplace (/vehicles):** Multi-faceted filter sidebar (Price, Brand, Year, Range, Battery, Body type, Condition, Location), Search bar, Natural Language query parser, Sort options, Pagination.
- **Vehicle Details (/vehicles/:id):** Gallery, spec sheet, range & battery breakdown, seller info, add to cart/wishlist/compare buttons, verified customer reviews.

- **Vehicle Comparison (/compare):** Up to 4 EVs comparison table with automatic badge highlights for Best Price, Best Range, Fastest Charging, Best Rating, Best Overall Value.
- **Wishlist & Cart Pages (/wishlist, /cart):** MongoDB backend synced cart & wishlist with stock validation & price calculations.
- **Checkout & Stripe Payment (/checkout):** Multi-step checkout with live Stripe Element / Payment Intent backend verification.
- **Order Tracking (/orders, /orders/:id/track):** Real-time visual timeline showing order progression from Placed to Delivered.

Phase 7: EV Tools & AI Recommendation Engine

- **EV Range & Charging Cost Calculators (/tools):** Real-time interactive calculations for daily usage, annual fuel savings vs ICE, charging costs.
- **AI Recommendation Engine (/recommendations):** Natural language + structured criteria input. System queries MongoDB candidates, ranks them using AI prompt (or intelligent fallback rule engine), and returns personalized matched EVs with rationale.

Phase 8: Vendor Portal (/vendor/*)

- **Vendor Dashboard:** MongoDB live metrics (Active listings, Revenue, Orders, Pending approvals, Rating), Sales chart with Recharts.
- **Listing Management:** Add EV listing with rich spec form & multi-image support, View/Edit/Delete listings, Mark Sold, track listing status (PENDING_APPROVAL, APPROVED, REJECTED).
- **Vendor Order Fulfillment:** Manage customer orders, update delivery status with timeline tracking.

Phase 9: Admin Portal (/admin/*)

- **Admin Dashboard:** Live system analytics, user growth, sales charts, revenue metrics, category distribution.
- **Listing Approvals:** Review pending EV submissions with Approve / Reject + Rejection reason workflow.
- **User & Vendor Management:** Manage user roles, suspend/activate vendors and users.
- **Order & Review Oversight:** Global order monitoring and review moderation.

Phase 10: Security, Performance & QA Polish

- Password hashing with bcrypt, JWT token expiry & authorization middleware.

- MongoDB query sanitization, CORS, Helmet security headers, rate limiting.
 - Responsive testing (320px to 1440px+), keyboard accessibility (ARIA), loading skeletons, error boundaries.
 - Full documentation (README.md, .env.example).
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Verification Plan

Automated & System Verification

1. cd backend && npm test or node-based API verification script.
2. Build checks for both backend and frontend (npm run build in frontend).
3. Full DB Seed check: verify 10+ realistic EVs, categories, sample users/vendors, and reviews in MongoDB.

Manual End-to-End Verification

- **Customer Flow:** Register/Login -> Browse Marketplace -> Natural Language Filter -> View Details -> Add to Compare -> Add to Cart -> Checkout & Payment -> Track Order -> Submit Review.
- **Vendor Flow:** Register Vendor -> Add EV Listing -> Check Pending Status -> Receive Order -> Update Order Status -> Check Revenue Analytics.
- **Admin Flow:** Login Admin -> Admin Dashboard -> Approve Pending Vendor EV -> Manage Users -> View Platform Analytics.
- **EV Intelligence:** Calculate Range & Charging Cost -> Run AI Vehicle Finder & verify results come from live MongoDB records.
- **UI Responsiveness & Theme:** Toggle Dark/Light mode, test on Mobile layout viewport.